



AI Playbook and Toolkit for Public Health Departments

Evaluating AI-Generated Health Education Materials

HED 130

This module teaches staff how to evaluate AI-generated or AI-assisted health education materials before they are used. It focuses on accuracy, source alignment, readability, accessibility, cultural relevance, equity, and fit for the intended audience. The module treats AI output as a draft that must be reviewed against explicit criteria before it becomes part of a training, handout, script, web page, or outreach product.

Level	applied
Primary track	Health Education Track
Appears in tracks	health-education

Learning Objectives

- Describe the review criteria needed for AI-generated health education materials.
- Identify common defects in AI-generated educational content, including factual errors, missing context, accessibility barriers, and stigma.
- Apply an evaluation rubric to an AI-assisted health education product.
- Create a revision plan that identifies what must be corrected before the material is used.

Module Overview

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Jurisdiction and Agency Policy Note

This national-level training module is intended for public health education, planning, and workforce development. It does not replace agency policy, legal review, privacy review, security review, procurement review, communications clearance, civil rights review, Tribal consultation, or governance approval. Learners should confirm applicable requirements, approved tools, data rules, and review pathways within their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

AI-generated health education materials can look polished while still being unsafe or ineffective. Fluent writing may hide inaccurate recommendations, missing contraindications, incorrect eligibility information, inaccessible formatting, or examples that do not fit the intended audience. Public health education must be actionable, accurate, equitable, accessible, and aligned with approved guidance. A structured evaluation rubric helps staff review materials consistently rather than relying on whether a draft sounds professional.

Evaluation is especially important when materials address clinical guidance, emergencies, vulnerable populations, community trust, or high-consequence decisions. AI may blend sources, overstate certainty, or create advice that sounds plausible but is not approved. It may also make materials too generic, leaving out local services, eligibility rules, or practical barriers. Reviewers need criteria that examine content, context, accessibility, and implementation feasibility.

Public Health Example

A program creates an AI-assisted handout about lead exposure prevention for families with young children. The draft includes accurate general concepts but omits local testing resources, uses technical terms such as exposure pathway, and includes an image prompt that shows only single-family homes even though many affected families live in rental housing. A reviewer uses a rubric to flag missing local resources, readability concerns, accessibility needs, and equity issues. The revised handout adds local contact information, plain-language explanations, tenant-focused examples, and an accessible layout.

Technical, Programmatic, and Operational Deep Dive

A strong evaluation rubric should begin with factual accuracy and source alignment. Reviewers should compare the AI-generated material to approved source documents, current program guidance, and local service details. They should check whether the draft introduces unsupported claims, omits limits, changes eligibility, or confuses

recommendations. When the source material is uncertain or evolving, the final material should explain uncertainty rather than presenting AI-generated certainty.

The rubric should also address readability, accessibility, and format. Readability includes sentence length, familiar words, clear headings, and direct action steps. Accessibility includes alternative text for images, logical reading order, adequate color contrast, meaningful link text, captioning or transcript needs for audio or video, and compatibility with assistive technology. Reviewers should also check whether the format works for the channel, such as print, mobile, social media, classroom training, or community events.

Equity and cultural relevance should be reviewed explicitly. Reviewers should ask who is represented, who is missing, whether examples reflect real living and working conditions, and whether the material acknowledges structural barriers. They should also examine whether the content places responsibility only on individuals when systems, resources, or policies affect health behavior. The goal is not to make every material address every social determinant, but to avoid inaccurate, stigmatizing, or impractical guidance.

Risks, Failure Modes, and Guardrails

A common failure mode is reviewing only grammar and reading level while missing factual or equity issues. AI-generated content may be grammatically strong and still be wrong, incomplete, or inappropriate. Guardrails include requiring source comparison, subject matter review, and equity/accessibility checks before use. Reviewers should document defects and revisions so the team learns what kinds of AI outputs require the most correction.

Another risk is allowing AI-generated materials to drift away from approved agency guidance. AI tools may generalize from broad knowledge and create statements that conflict with local programs, funding rules, emergency instructions, or legal requirements. Reviewers should require the material to identify its source base and should remove unsupported claims. When no approved source exists, the material should be routed for subject matter review before use.

A third risk is treating evaluation as a final publication step rather than an iterative process. If reviewers consistently find the same problems, the team may need better prompts, clearer source packets, different workflows, or a decision not to use AI for that content type. Evaluation findings should inform training, governance, and tool-use guidance. This turns review into quality improvement rather than a one-time correction.

Application to AI-Supported Workflows

This module applies to workflows where AI drafts educational materials, training scenarios, scripts, FAQs, handouts, slide text, or web content. The workflow should require review criteria before the first draft is generated so staff know what quality means. Reviewers should record whether the material is ready to use, needs revision, or should not be used. For high-risk topics, the evaluation should include subject matter, communications, accessibility, equity, legal, or program review as required by agency policy.

Reflection Questions

What review criteria are currently used for educational materials in your organization?

What AI-generated content defects would create the most risk for your audience?

How should repeated AI output problems be documented and used to improve the workflow?

Practical Exercise

Create a rubric for evaluating one AI-generated health education material. Include criteria for source alignment, factual accuracy, actionability, readability, accessibility, cultural relevance, stigma, local context, and required reviewers.

Example: For an AI-assisted food safety handout, the rubric might require that cooking temperature guidance matches approved sources, instructions are understandable without technical training, images include alt text, examples include renters and shared kitchens, and final review is completed by environmental health and communications staff.

Expected Artifact or Evidence

Evaluation rubric for AI-generated health education material.

Short revision plan identifying at least three likely corrections.

Human review pathway for final clearance.

Knowledge Check

- 1. Why is grammar review insufficient for AI-generated educational materials?
- 2. What does source alignment mean?
- 3. Which review area addresses screen-reader compatibility and alt text?
- 4. What should happen when an AI draft repeatedly creates the same problem?
- 5. Which rating is most appropriate for formative review?

References and Resources

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